IND209

concepts and interactions

Prototyping interaction

interaction *introduction*

reciprocal...

interaction

/ˌɪntərˈakʃ(ə)n/

noun

reciprocal action or influence.
"ongoing **interaction between** the two languages"

Sinonimi: interplay | interchange | interactivity | interconnection | interlinkage

- communication or direct involvement with someone or something.
  "for a shy person, social interaction can be a stomach-churning, anxiety-filled experience"

  Sinonimi: contact | relations | connection | association | communion

- PHYSICS
  a particular way in which matter, fields, and atomic and subatomic particles affect one another, e.g. through gravitation or electromagnetism.

https://translate.google.com

interaction *introduction*

5 dimensions of Interaction Design

1D: Words *e.g., button labels—should be meaningful and simple to understand.*

2D: Visual representations *Graphical elements like images, typography and icons that users interact with.*

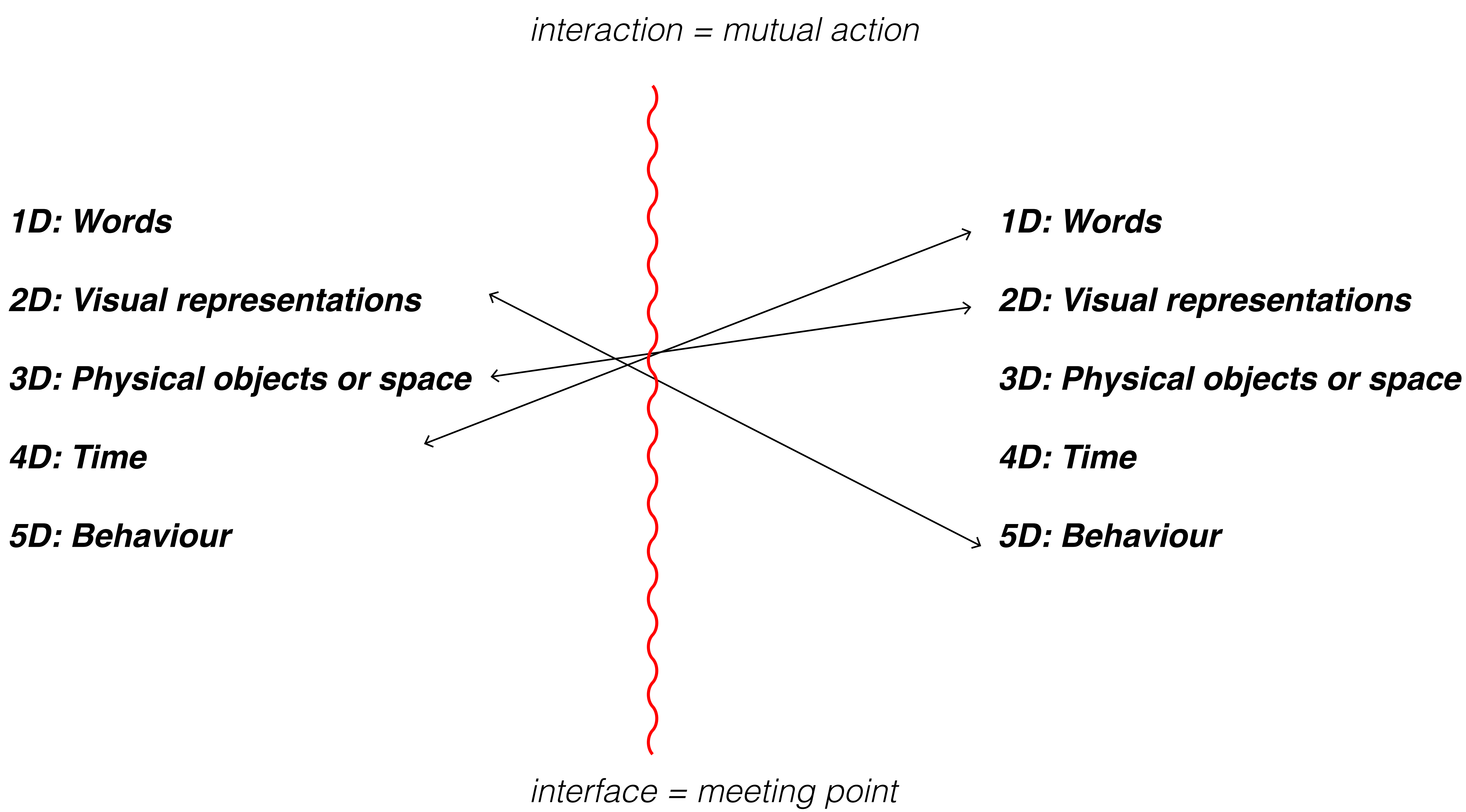
3D: Physical objects or space *Through what physical objects do users interact with the product? And within what kind of physical space does the user do so?*

4D: Time *The amount of time a user spends interacting with the product.*

5D: Behaviour *How do users perform actions and operate the product?*

Teo, Y. S. (2025, March 15). What is Interaction Design?. Interaction Design Foundation - IxDF. https://www.interaction-design.org/literature/article/what-is-interaction-design

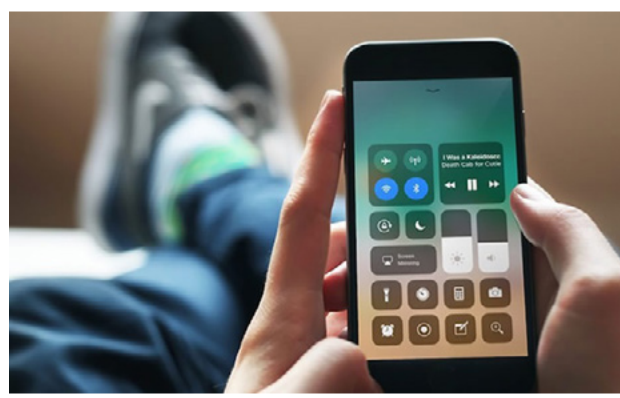
interaction *introduction*



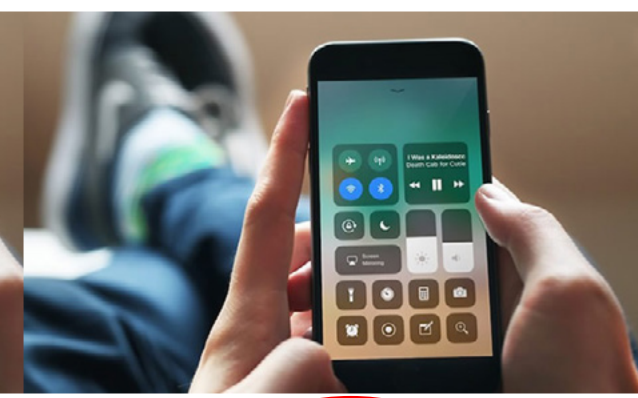
interaction *introduction*

Types of interfaces:

graphic

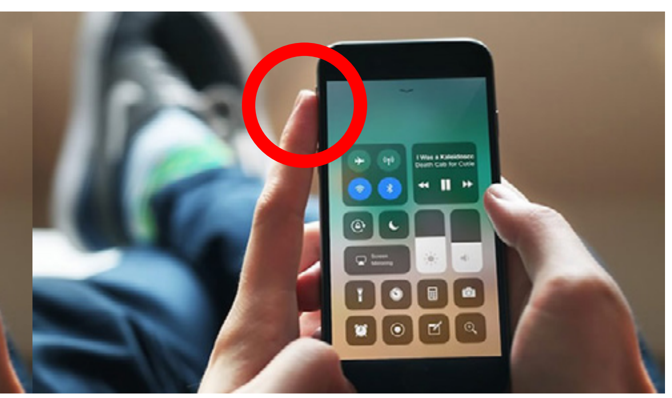


voice



Hey Siri!

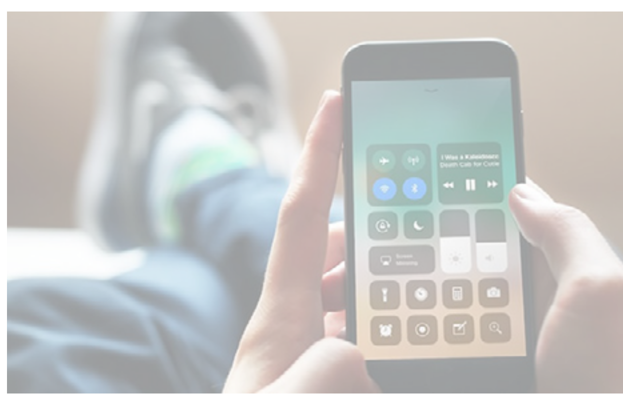
tangible



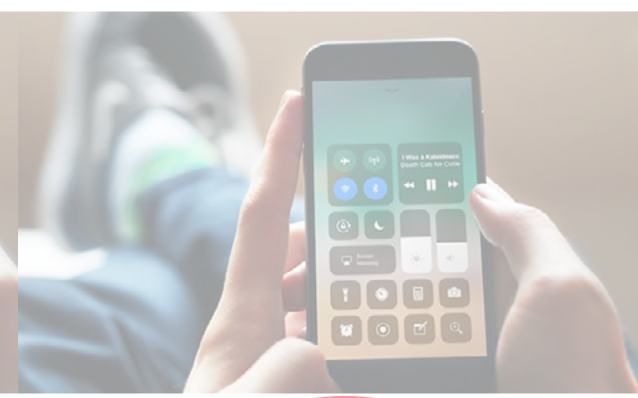
interaction *introduction*

Types of interfaces:

graphic



voice



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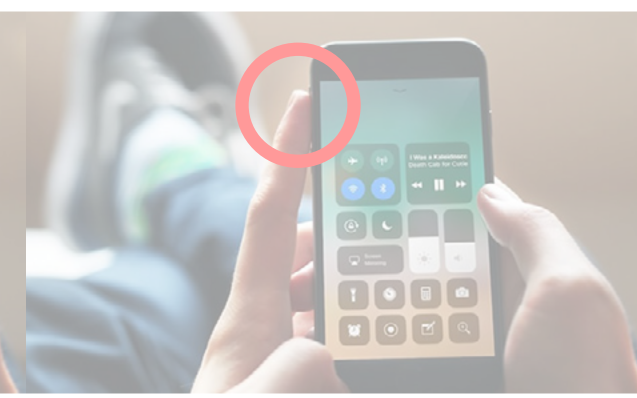
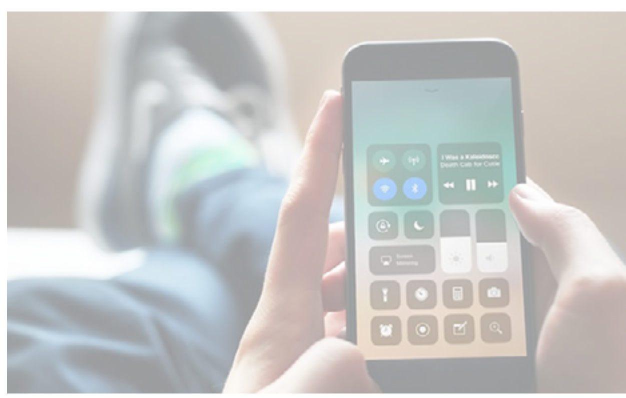


service

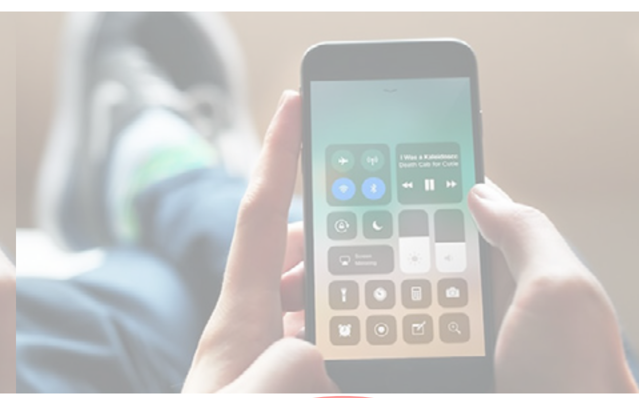
interaction *introduction*

Types of interfaces:

graphic

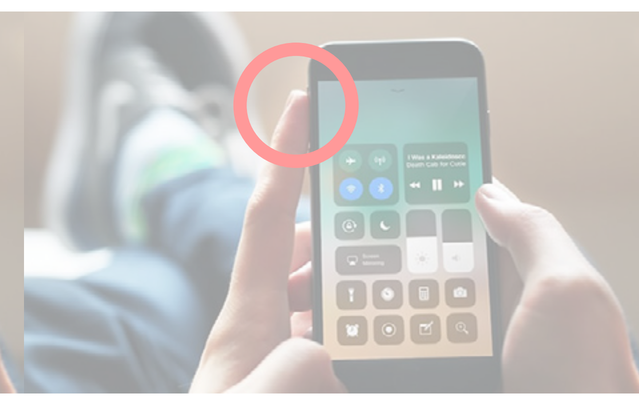


voice



Hey Siri!

tangible



service

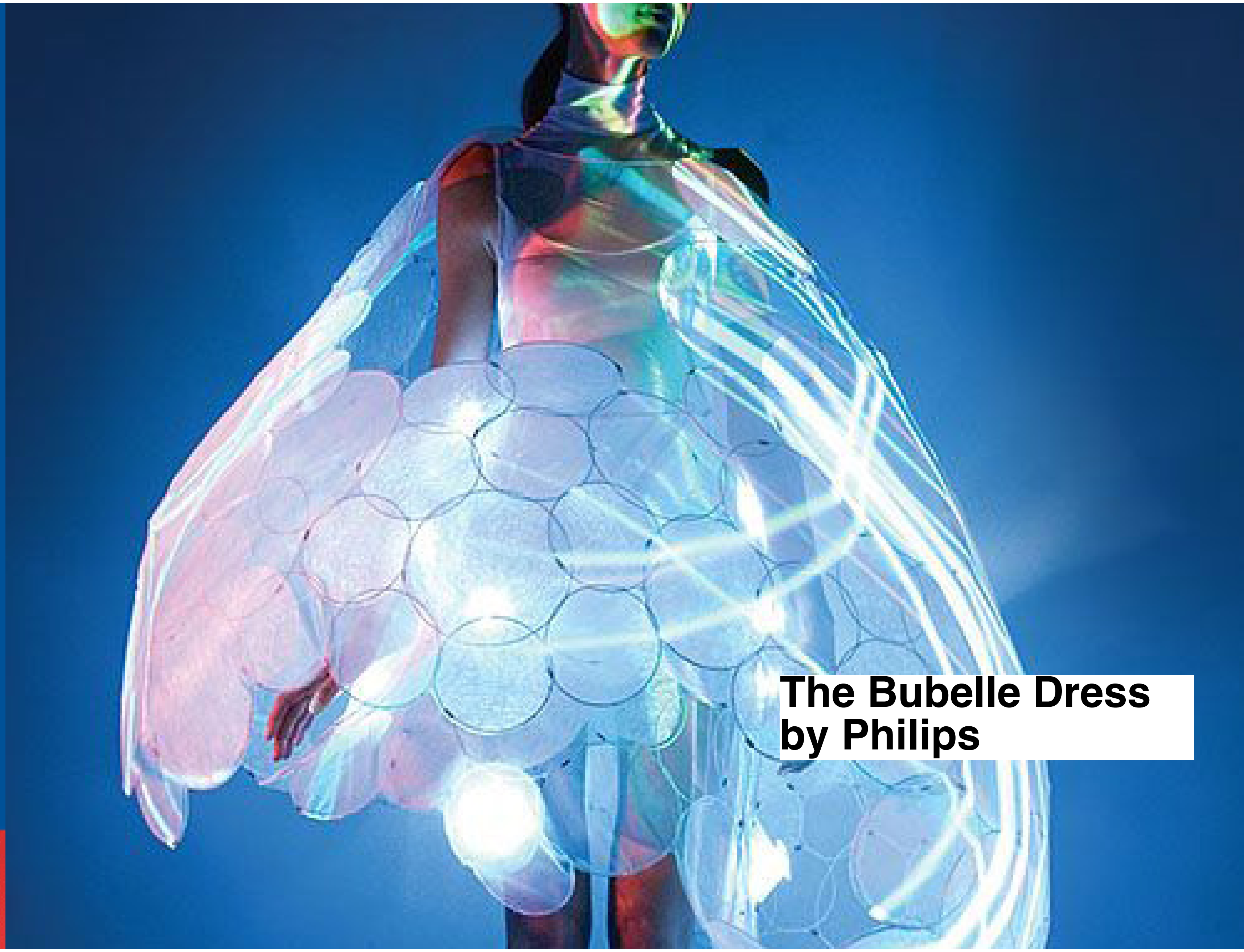
spatial and virtual

interaction *introduction*

Who/what interacts?

human interacts with: human, object, surface, interface, environment, software, etc.
object: collects data, responds to inputs

Amazon Echo



The Bubelle Dress by Philips

When designing interaction *consider*

Cognitive interaction components:

what is the **goal** of interaction?

Which **elements** are involved in interaction (buttons, surfaces, etc.)?

what is **perceived** (sounds, symbols, etc)?
and what is the eventual user **action**?

When designing interaction *consider*

Affordances:
How will your user understand what are the
possible actions *with your product?*
Which elements will help your users perceive possibilities for interaction?

Superpatata by Hector Serrano, 2000.

Volé lamp by Mauricio Sanin & Kelly Durango Cardenas, 2018.

When designing interaction *consider*

Symbolic interactionism:
*People live in natural and **symbolic environment** ...*
*it emphasizes **subjective understanding** of symbols such as words, definitions, roles, gestures, rituals, etc.*

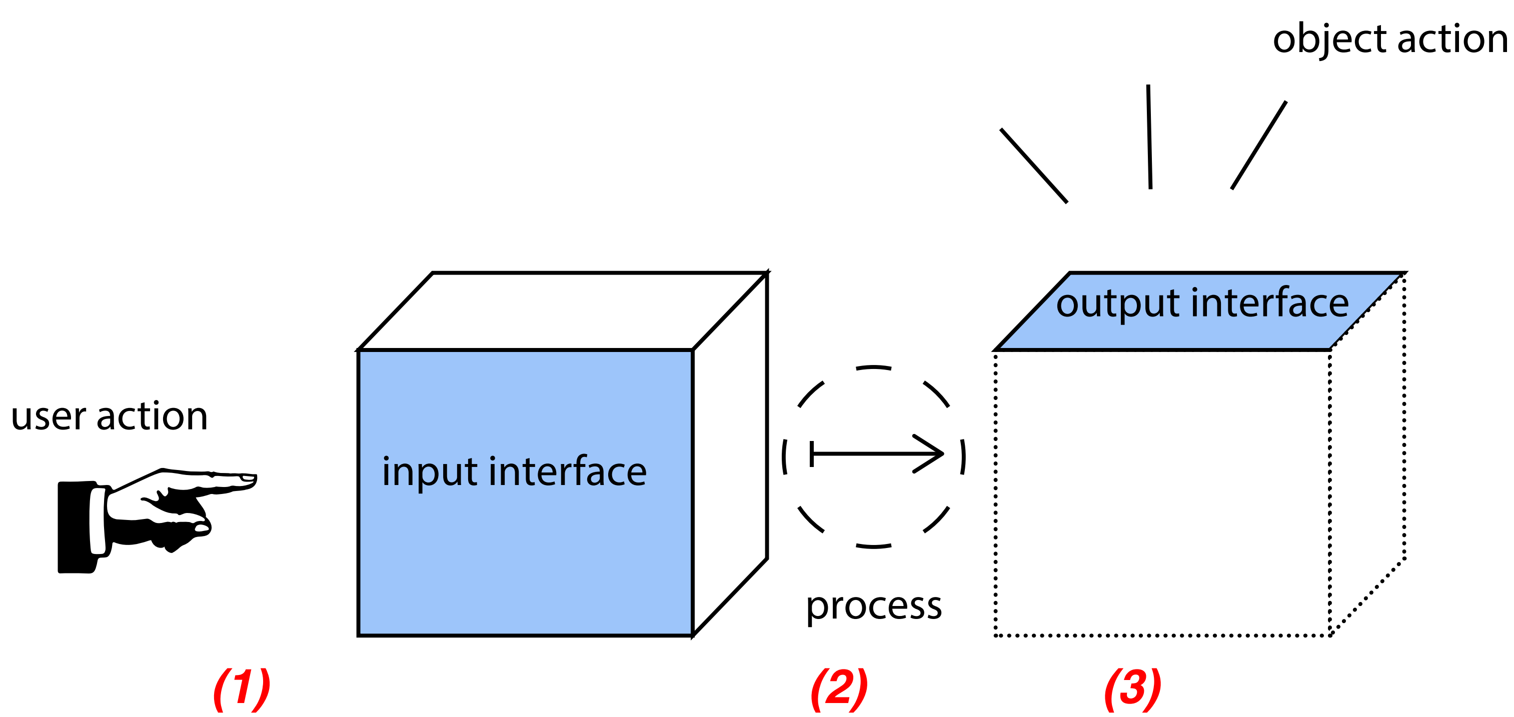
Mead, G. H. (1934). Mind, self and society.

When designing interaction *consider*

Decomposing your project:

- *categories of shapes*
- *list of materials*
- *list of gestures*
- *list of inputs and outputs*

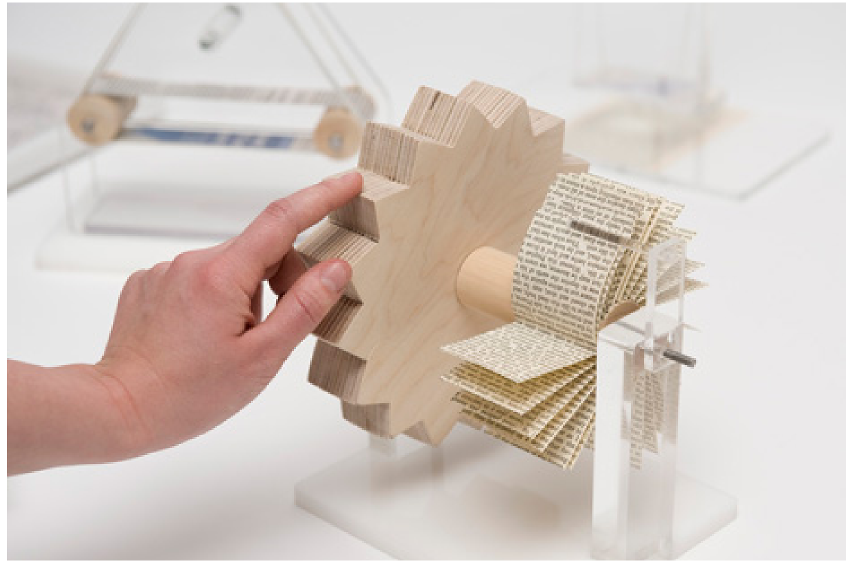
gestures	input	output	shapes	materials



Designing *alternative interactions*

Multi-Touch Gestures by Gabriele Meldaikyte:
Royal College of Art student Gabriele Meldaikyte has designed a set of interactive exhibits for a museum of iPhone gestures.

pinch gesture



flick gesture

scroll gesture

swipe gesture

tap gesture

Designing *alternative interactions*

Substitute Phone by Klemens Schillinger:
This product offers a therapeutic appropach,
offering help to smartphone addicted.

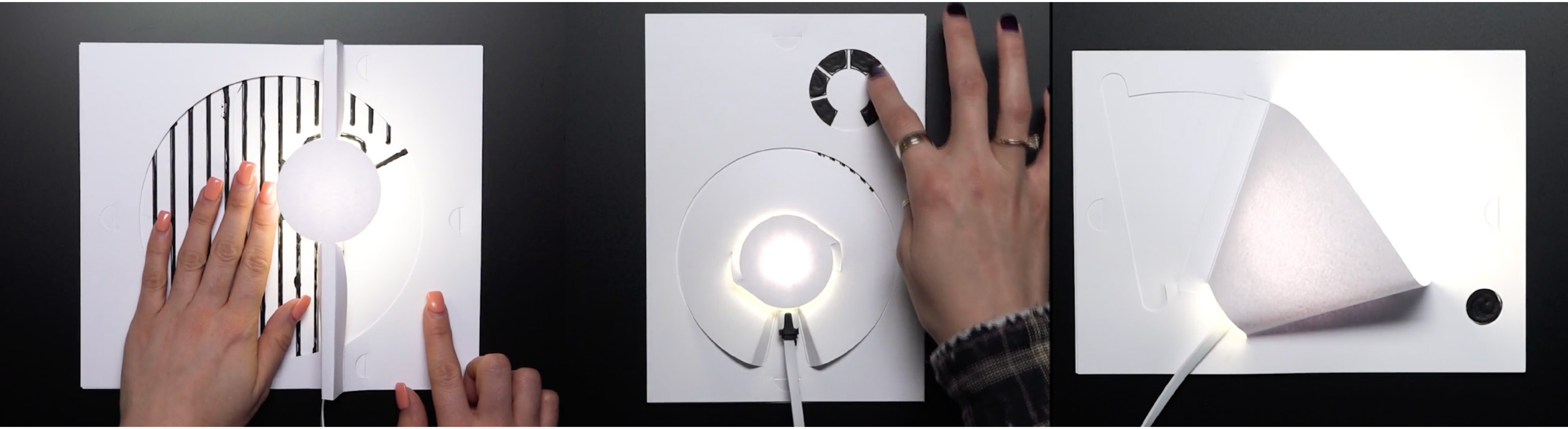
Designing *alternative interactions*

Cord UIs MIT labs

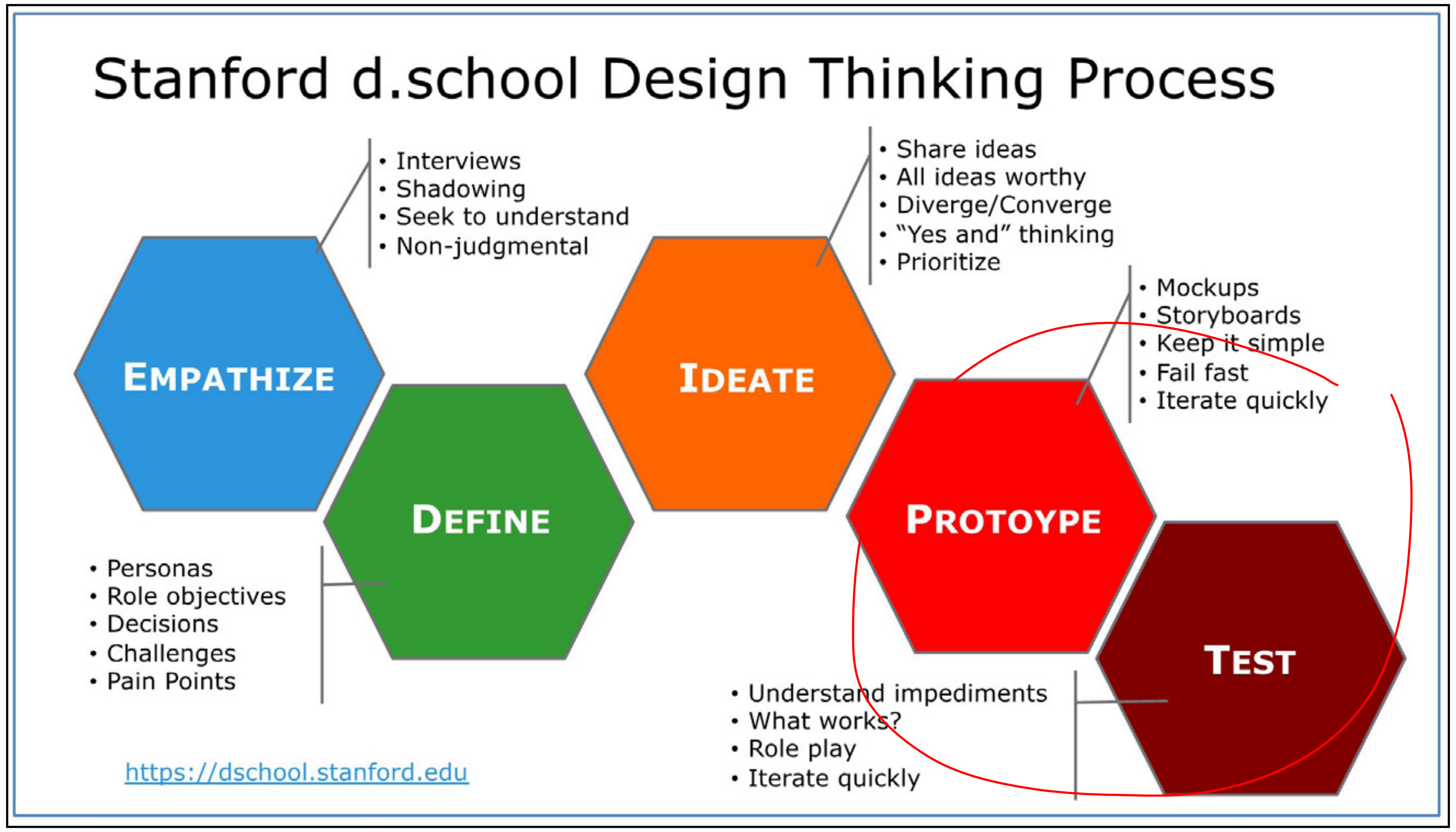
Cord UIs are sensorial augmented cords that allow for simple metaphor-rich interactions to interface with their connected devices.

Designing *alternative interactions*

Bare Conductive Electric Paint Lamp Kit
Transform paper into light! This Electric Paint Lamp Kit gives you the choice of three uniquely cool paper lamps.



Design process *design phases*



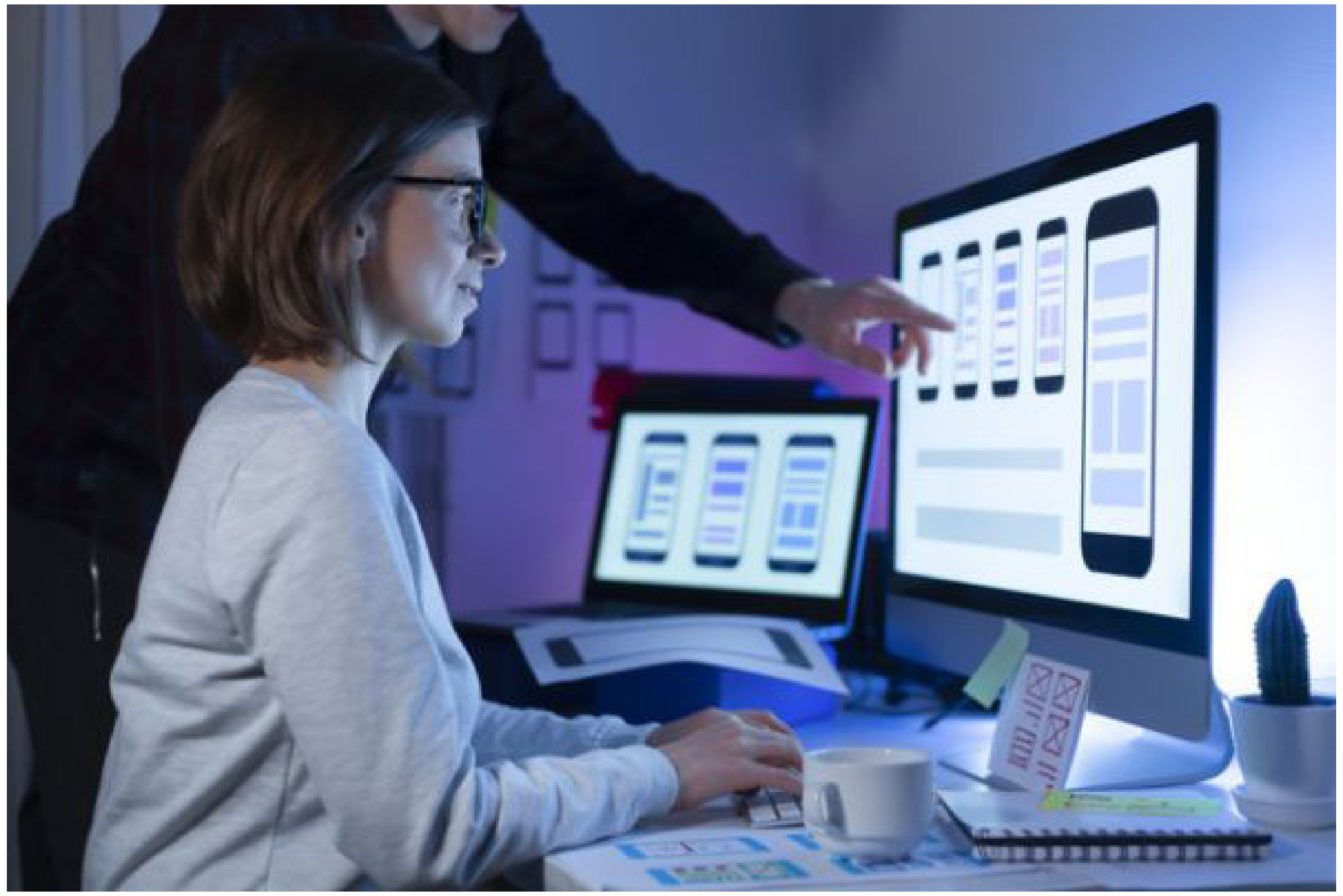
Design process *User Testing*

User Testing *is a research method where real users interact with a product (website, app, or prototype) to evaluate its ease of use, functionality, and overall experience.*
The goal is to identify problems before launch and improve design decisions based on real feedback.

What can be validated?

USABILITY (ease of use)
Ergonomics
Usefulness
Effectiveness
Learning ability
Error prevention

USER EXPERIENCE
Pleasure
Satisfaction
Interaction
Emotions



Design process *User Testing*

How it can be reported?

USABILITY FEEDBACK
“user spent less time on this task with this version of design”

USER EXPERIENCE FEEDBACK
“most users found this design solution as expressing warm feelings, which corresponds with the expected results”

Design process *User Testing*

key steps

step 1: ***use cases***

How it can be used?

Which functions it has?

Where it will be used?

step 2: ***users***

Who are your users?

Consider extremely different users ...

What is the minimum amount of users you need for the test?

Consider ethics ...

step 3: ***tasks***

Based on step 1, repare a list of tasks that users will perform, e.g. “Log in”, “Find xyz”, etc.

step 4: ***plan***

Which methods are the most suitable?

Which tools will you need?

Which wireframes should you prepare?

How much time it will take to conduct a test?

step 5: ***execute***

Instruct the user according to the plan and conduct the test.

Document the test execution.

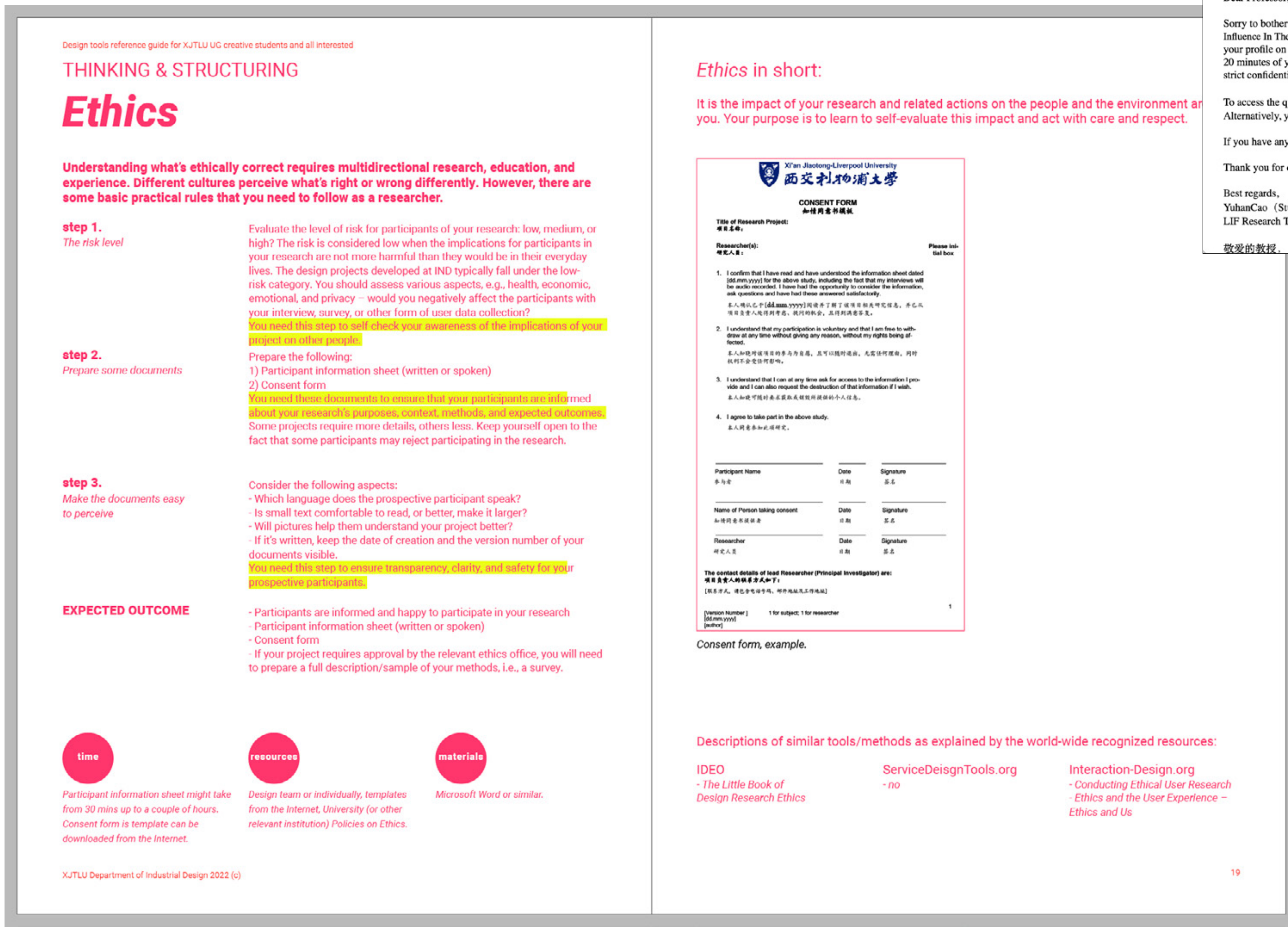
step 6: ***analyze***

What are the defects of the design?

What are the strong sides of your design?

Design process *User Testing*

before you start



Design tools reference guide for XJTLU UG creative students and all interested

THINKING & STRUCTURING

Ethics

Understanding what's ethically correct requires multidirectional research, education, and experience. Different cultures perceive what's right or wrong differently. However, there are some basic practical rules that you need to follow as a researcher.

step 1. *The risk level*

Evaluate the level of risk for participants of your research: low, medium, or high? The risk is considered low when the implications for participants in your research are not more harmful than they would be in their everyday lives. The design projects developed at IND typically fall under the low-risk category. You should assess various aspects, e.g., health, economic, emotional, and privacy – would you negatively affect the participants with your interview, survey, or other form of user data collection? You need this step to self-check your awareness of the implications of your project on other people.

step 2. *Prepare some documents*

Prepare the following:
1) Participant information sheet (written or spoken)
2) Consent form
You need these documents to ensure that your participants are informed about your research's purposes, context, methods, and expected outcomes. Some projects require more details, others less. Keep yourself open to the fact that some participants may reject participating in the research.

step 3. *Make the documents easy to perceive*

Consider the following aspects:
- Which language does the prospective participant speak?
- Is small text comfortable to read, or better, make it larger?
- Will pictures help them understand your project better?
- If it's written, keep the date of creation and the version number of your documents visible.

You need this step to ensure transparency, clarity, and safety for your prospective participants.

EXPECTED OUTCOME

- Participants are informed and happy to participate in your research
- Participant information sheet (written or spoken)
- Consent form
- If your project requires approval by the relevant ethics office, you will need to prepare a full description/sample of your methods, i.e., a survey.

time — *Participant information sheet might take from 30 mins up to a couple of hours. Consent form is template can be downloaded from the Internet.*

resources — *Design team or individually, templates from the Internet, University (or other relevant institution) Policies on Ethics.*

materials — *Microsoft Word or similar.*

XJTLU Department of Industrial Design 2022 (c)

Ethics in short:

It is the impact of your research and related actions on the people and the environment ar[ound] you. Your purpose is to learn to self-evaluate this impact and act with care and respect.

Xi'an Jiaotong-Liverpool University
西交利物浦大学

CONSENT FORM
知情同意书模板

Title of Research Project:

Researcher(s):

Please initial box

1. I confirm that I have read and have understood the information sheet dated [dd.mm.yyyy] for the above study, including the fact that my interviews will be audio recorded. I have had the opportunity to consider the information, ask questions and have had these answered satisfactorily.
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason, without my rights being affected.
3. I understand that I can at any time ask for access to the information I provide and I can also request the destruction of that information if I wish.
4. I agree to take part in the above study.

Participant Name	Date	Signature
Name of Person taking consent	Date	Signature
Researcher	Date	Signature

The contact details of lead Researcher (Principal Investigator) are:

[Version Number] [dd.mm.yyyy] [author] 1 for subject; 1 for researcher 1

Consent form, example.

Descriptions of similar tools/methods as explained by the world-wide recognized resources:

IDEO
- The Little Book of Design Research Ethics

ServiceDeisgnTools.org
- no

Interaction-Design.org
- Conducting Ethical User Research
- Ethics and the User Experience – Ethics and Us

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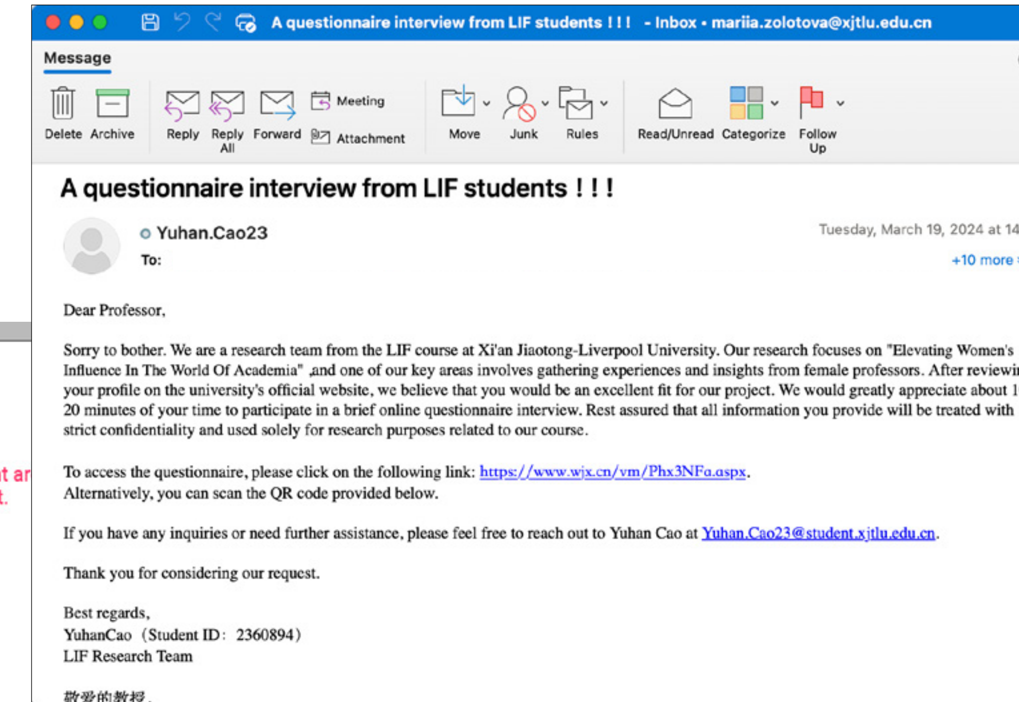


A questionnaire interview from LIF students ! ! ! - Inbox • mariia.zolotova@xjtlu.edu.cn

A questionnaire interview from LIF students ! ! !

Yuhan.Cao23 — Tuesday, March 19, 2024 at 14:54

To: +10 more

Dear Professor,

Sorry to bother. We are a research team from the LIF course at Xi'an Jiaotong-Liverpool University. Our research focuses on "Elevating Women's Influence In The World Of Academia" ,and one of our key areas involves gathering experiences and insights from female professors. After reviewing your profile on the university's official website, we believe that you would be an excellent fit for our project. We would greatly appreciate about 10-20 minutes of your time to participate in a brief online questionnaire interview. Rest assured that all information you provide will be treated with strict confidentiality and used solely for research purposes related to our course.

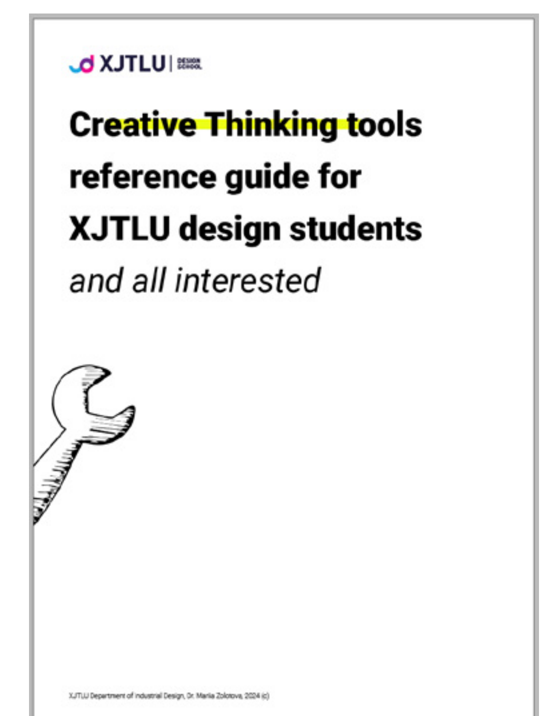
To access the questionnaire, please click on the following link: https://www.wjx.cn/vm/Phx3NFa.aspx.
Alternatively, you can scan the QR code provided below.

If you have any inquiries or need further assistance, please feel free to reach out to Yuhan Cao at Yuhan.Cao23@student.xjtlu.edu.cn.

Thank you for considering our request.

Best regards,
YuhanCao（Student ID： 2360894）
LIF Research Team

敬爱的教授，



https://figshare.com/articles/preprint/Creative_Thinking_tools_reference_guide_for_XJTLU_UG_creative_students_and_all_interested/25602636/4

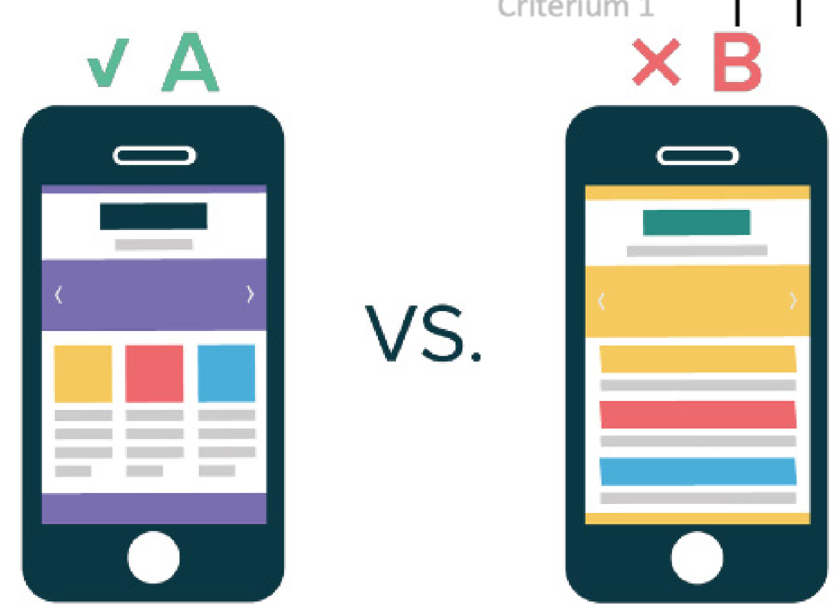
Design process *User Testing*

Testing different options

A/B tests*: what can be tested?*

- Information Architecture
- User Interface
Smaller details, such as
- Headlines
- Calls to action, text/ location
- Pop-up windows
- Images
- The number of fields in the form
- The color of the button or the surrounding space
etc.

When planning your tests, you can think:
"I expect [the design option A] to solve [the problem]/ help user achieve [result]."



User 1. **prototype A (image)**

	--	-	+	++
Criterium 1				
Criterium 2				
Criterium 3				
Criterium 4				
Criterium 5				
Criterium 6				

User 1. **prototype B (image)**

	--	-	+	++
Criterium 1				
Criterium 2				
Criterium 3				
Criterium 4				
Criterium 5				
Criterium 6				

User 2. **prototype A (image)**

	--	-	+	++
Criterium 1				

User 2. **prototype B (image)**

	--	-	+	++
Criterium 1				
Criterium 2				
Criterium 3				
Criterium 4				
Criterium 5				
Criterium 6				

Design process *User Testing*

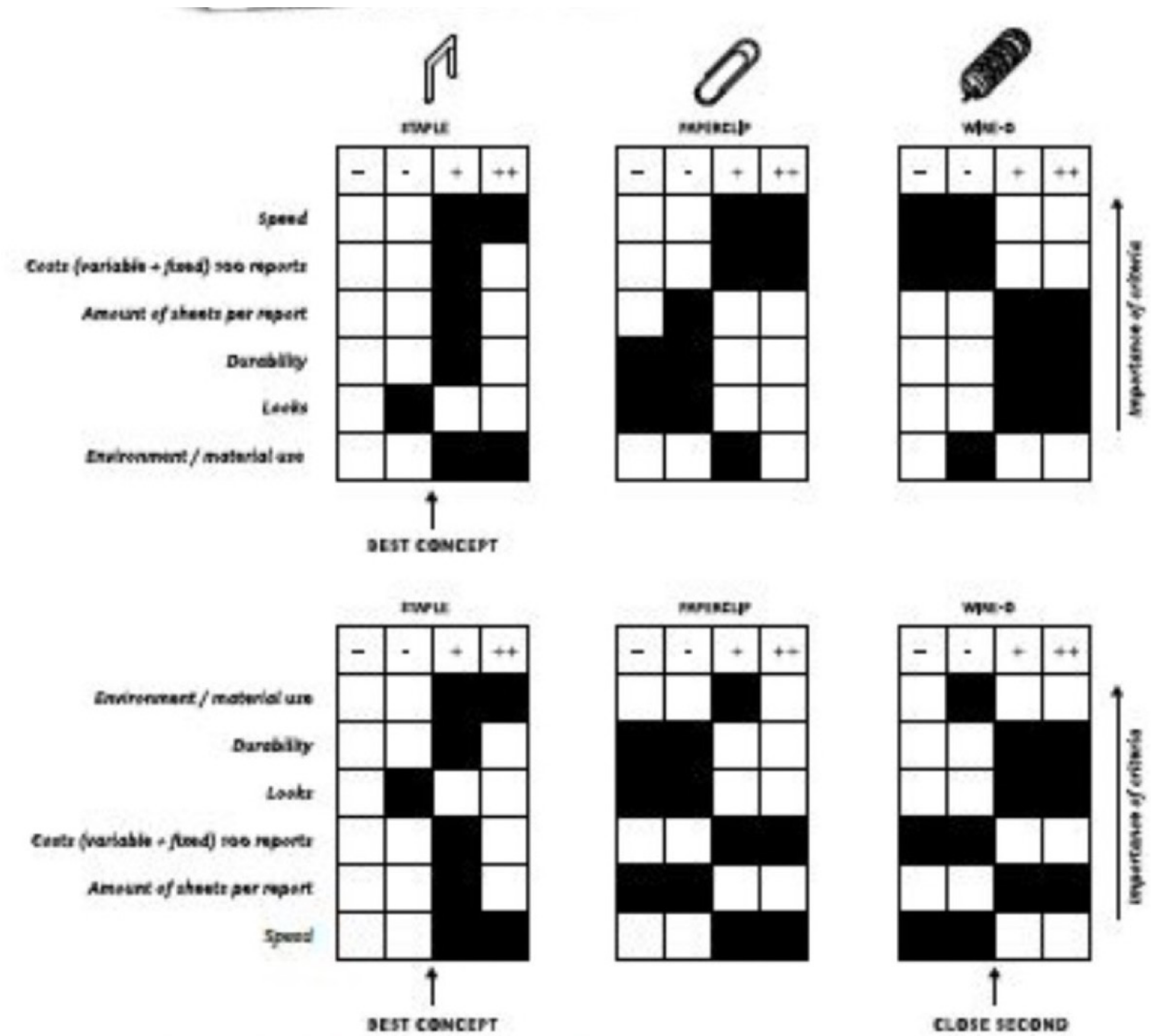
***A/B tests**: how to conduct?*

1. Identify and list the requirements specific to a successful design concept.

2. Next to the list of requirements, create a matrix as in the example on the right.

3. Ask the user to interact with each design option and evaluate it according to the criteria.

4. Compare the results and choose the most promising one.



Design process *User Testing*

Testing user flow / wireframes / UI

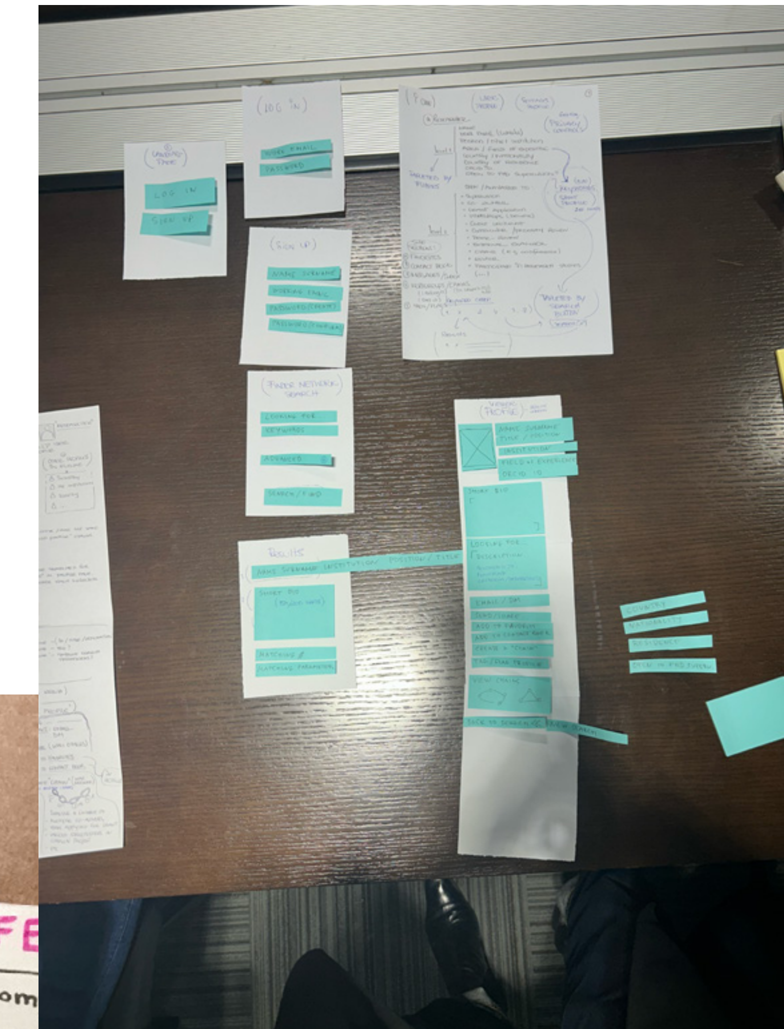
Paper (Figma) Prototypes*: how to conduct?*

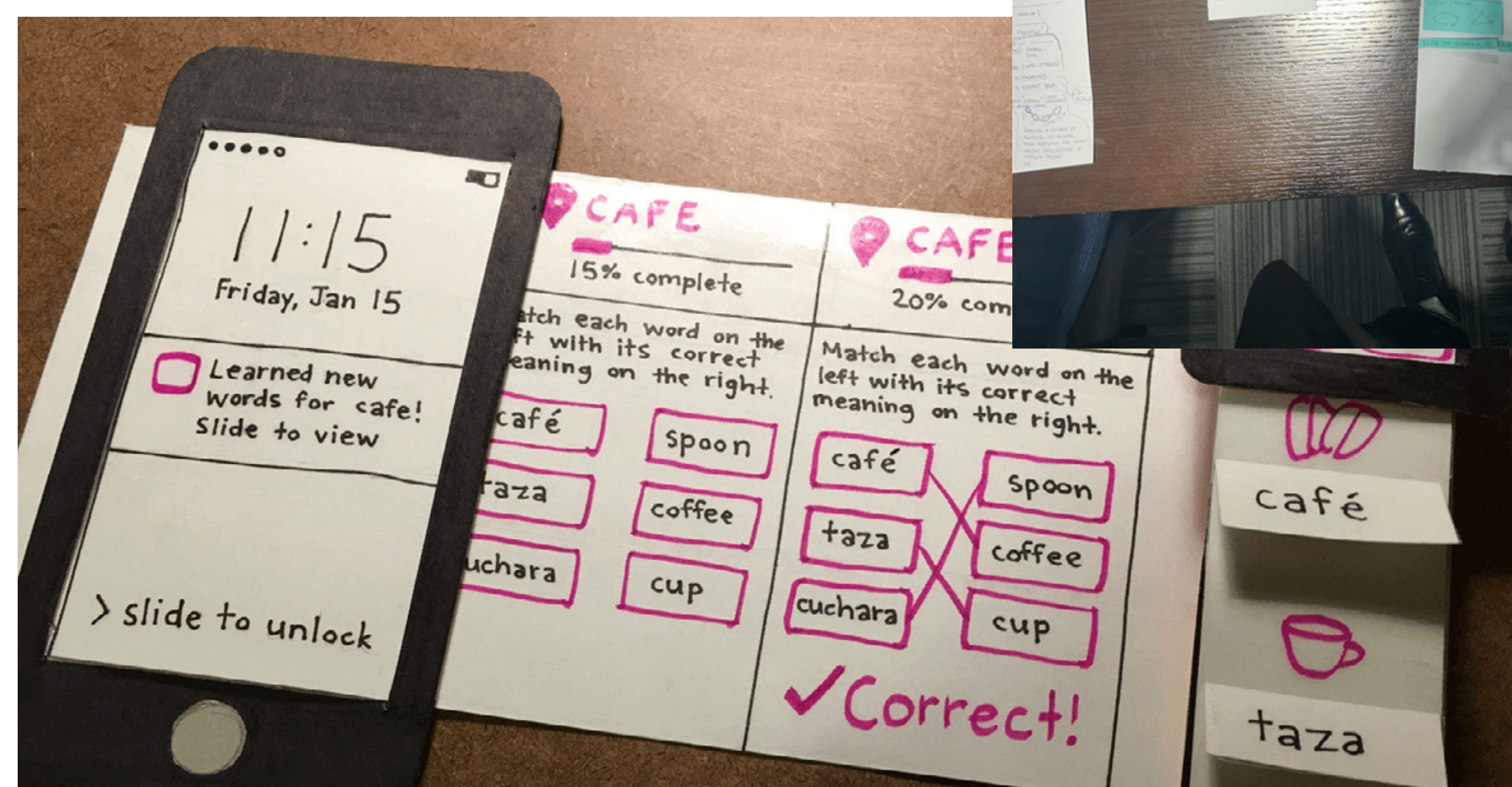
1. Determine which features/interaction steps you would like to test.

2. Design a paper interface that matches your request.

3. Ask the user to complete the tasks of interacting with the interface, but do not tell them how to do it correctly, observe how easy and difficult it is for the user to cope with the task.

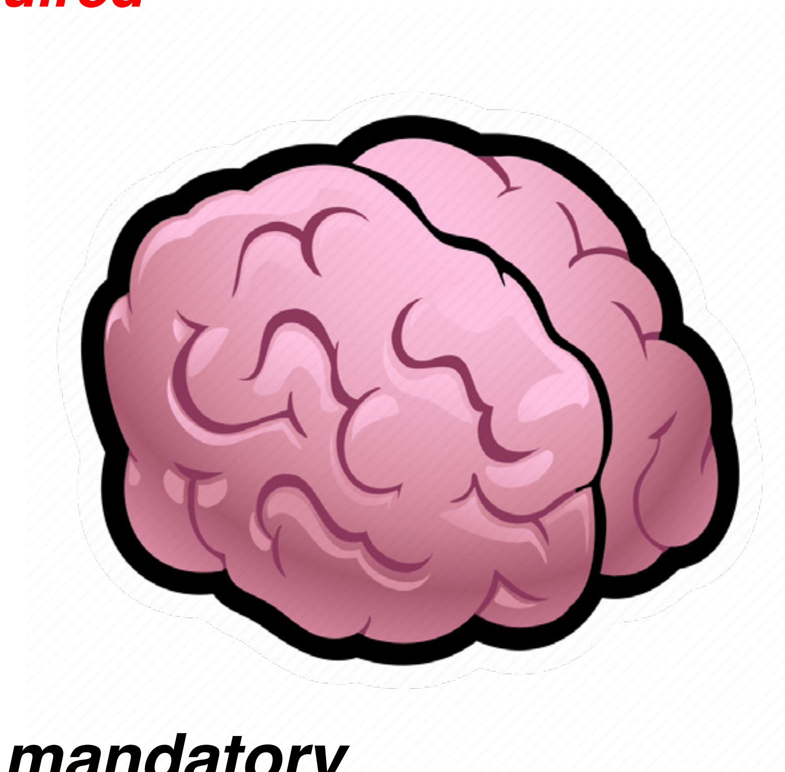
>> https://www.nngroup.com/articles/paper-prototyping/





Design process *User Testing*

tools required



mandatory

Figma

optional ...

Thank you! *Questions?*